Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Develop Multi AI Agent Applications with Gemini Agent ADK

TGS Ref No: TGS-2024042961

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.3

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	12 August 2026	Initial release — 2-day lesson plan for Develop Multi AI Agent Applications with Gemini Agent ADK, aligned to the 18 hands-on labs across the four course topics.	Dr Alfred Ang
1.1	12 August 2026	QA fixes — aligned the Practical Performance duration to 1 hour across all artifacts.	Dr Alfred Ang
1.2	11 August 2026	Restructured the labs into one self-contained folder per lab (lab01 … lab18); lab references updated throughout.	Dr Alfred Ang
1.3	12 August 2026	Slide deck rebuilt to the expanded WSQ visual standard: each lab now runs briefing → process map → procedure (with commands) → verify with troubleshooting; added staged process maps with real connectors, decision maps, comparison matrices, worked examples and native charts; restrained slide transitions applied throughout.	Dr Alfred Ang

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Course Information

Course Title	Develop Multi AI Agent Applications with Gemini Agent ADK
WSQ Course Reference	TGS-2024042961
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on agent-development labs using Python, the Google Agent Development Kit and Gemini
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyze the range of LLM applications using Generative AI (GAI) and identify their industrial use cases.
- LO2: Establish Google Gemini GAI designs and assess improvements on engineering processes.
- LO3: Develop LLM applications and assess its feasibility.
- LO4: Evaluate the performance effectiveness of Retrieval Augmented Generation (RAG).

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on ADK agent build tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted at the end of Day 2.
- A minimum of 75% attendance is required to be eligible for assessment and funding. Open book covers the slides, Learner Guide and approved materials only.

Course Schedule

Day 1 — Agentic AI foundations with Gemini ADK — first agents, tools, sessions and multi-agent handoff

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, learning outcomes, ground rules and mandatory digital attendance (AM)
10:00–11:15	75 min	**Topic 1 — Overview of Agentic AI in Gemini ADK: what an LLM is, LLM use**

		cases across industries, agentic AI, the Gemini model family and ADK architecture (concepts + demo)
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 1: Set Up the Gemini ADK Environment and Get an API Key; Lab 2: Build Your First ADK Agent — A Retail Banking Assistant
13:00–14:00	60 min	Lunch break. Mandatory digital attendance (PM) on return
14:00–15:45	105 min	Hands-on: Lab 3: Give an Agent Tools — Live Weather and Web Search; Lab 4: Swap the Model — Running an ADK Agent on a Non-Gemini LLM
15:45–16:00	15 min	Tea break
16:00–16:45	45 min	**Topic 2 — Build A Multi Agent App with Gemini ADK: sessions and state, the Runner, the event loop and multi-agent handoff (concepts + demo)**
16:45–18:15	90 min	Hands-on: Lab 5: Give an Agent Memory — Sessions, State and the Runner; Lab 6: Inspect the Agent Loop — Events, Tool Calls and Final Responses; Lab 7: Multi-Agent Handoff — Joke Generator to Translator; Lab 8: Hierarchical Multi-Agent System — The Tutor Agent
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Multi-agent systems, agentic RAG and shipping an agent app with Streamlit

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	Topic 2 continues — workflow agents, guardrails, structured output and MCP (concepts). Hands-on: Lab 9: Sequential Workflow Agent — Singapore Transport Route Planner; Lab 10: Add a Guardrail — Blocking Unsafe Requests with a Callback
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	Hands-on: Lab 11: Structured Output — Forcing Valid JSON with Pydantic; Lab 12: Connect External Tools with MCP —

		StreamableHTTP and SSE
13:00–14:00	60 min	Lunch break. Mandatory digital attendance (PM) on return
14:00–15:15	75 min	Topic 3 — Build Agentic AI RAG in Gemini ADK: the RAG pipeline, chunking, embeddings, vector stores and retrieval (concepts). Hands-on: Lab 13: Load, Split and Embed Documents into a Vector Store; Lab 14: Build the Agentic RAG Agent — Retrieval as a Tool; Lab 15: Evaluate RAG Performance — Retrieval Quality and Groundedness
15:15–15:30	15 min	Tea break
15:30–16:00	30 min	Topic 4 — Agentic AI App with Gemini ADK and Streamlit (concepts). Hands-on: Lab 16: Declarative Agents — Configuring a Multi-Agent System in YAML; Lab 17: Ship the Agent as a Web App with Streamlit; Lab 18: Capstone — Design, Build and Assess Your Own Multi-Agent Application
16:00–16:15	15 min	Course recap, Q&A, TRAQOM survey and Assessment digital attendance
16:15–16:30	15 min	**Briefing for Assessment**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — hands-on Gemini ADK agent build tasks, 1 hour, open book**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to course topics)

Topic / Skill area	Weighting	Labs
Topic 01: Overview of Agentic AI in Gemini ADK	25%	Lab 1, Lab 2, Lab 3, Lab 4
Topic 02: Build A Multi Agent App with Gemini ADK	35%	Lab 5, Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12
Topic 03: Build Agentic AI RAG in Gemini ADK	25%	Lab 13, Lab 14, Lab 15
Topic 04: Build an Agentic AI App with Gemini Agent ADK and Streamlit	15%	Lab 16, Lab 17, Lab 18